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ITA-1408 Ethical Hacking

ASSESSMENT Tool - 1
C03

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Assessment Tool-3

1) Nmap Scan Without Permission:

Analysis:

The Students actions are non Ethical may be illegal. Although Nmap is a Legitimate network Scanning tool, using it Against another Organ network without prior authorization violates ethical Hacking Principles

Ethical Hacking Requires:

- write Permission From the System Owner
- Clearly defined Scope
- Responsible reporting

Scanning a Company's network without Consent Can be Considered as an unauthorized attempt to Access Computer resources..

Relevant Provisions of the IT Act, 2000

Section 43:

- Unauthorized Access or Interference with Computer Systems.
- The organization may Claim Compensation For damages.

Section 66:

- IF the act under - Section 43 is performed dishonestly or Fraudulently it becomes a Criminal offense.
- Punishment may include imprisonment and / or Fine..

Section 72:

- Covers breach of Confidentially and Privacy if Combined information is Accessed...

Possible Legal Consequences:

- Police Complaint or cybercrime investigation
- Compensation under Section 43
- Criminal prosecution under Section 66
- Damage to Professional reputation.

Correct Procedure:

The Student Should Leave:

- 1.) Obtained written authorization
- 2.) Defined the Scope of testing
- 3.) Signed a Legal agreement
- 4.) Conducted only Approved Security tests.
- 5.) Submitted a Confidential Security reports.
- 6.) Avoided Scanning Systems outside that Approved Scope...

Conclusion:-

Even if no damage occurred Scanning without Permission is unethical and may violate the Information Technology Act 2008

2) SQL Injection Vulnerability Published on Social media...

A) Analysis:

The researcher's actions are unethical because public disclosure occurred before the organization had an opportunity to fix...

Responsible Vulnerability Disclosure:-

- Privately inform the organization
- Providing sufficient technical details
- Allowing adequate time for remediation
- Publishing only after the vulnerability or agreed disclosure has expired

Risk of Premature Disclosure:-

- Attackers exploit the vulnerability immediately
- Customer information is compromised.
- Financial Loss.
- Loss of customer trust
- Legal liability

Relevant IT Act Provisions:-

Section 43:-

- unauthorized Access causing damage.

Section 66:-

- Criminal liability for dishonest fraudulent-related offenses.

Section 72

- Disclosure of Confidential information with authorization.

Conclusion:-

Responsible vulnerability disclosure protects both users and organization while promoting cybersecurity...

3) Employee Falls For Fake IT Support Email:-

Type of Cyber Attack:-

This is Phishing Attack (Credential phishing involving Social Engineering)

Attack Process:-

- 1) Attacker Sends a Fake Email
- 2) Employee believe it is Legitimate
- 3) Employee enters Credentials
- 4) Credentials are Stolen.
- 5) Attacker gains unauthorized access
- 6) Confidential Company information is Compromised..

Legal Implication under the ICT Act.

Section 43: Unauthorized Access to Computer Systems

Section 66: Computer-related offenses committed dishonestly.

Section 66C: Identify theft using Stolen Logical Credentials.

Section 66D: Cheating by personation using Computer Resources...

Responsibilities of the Organization:-

- Employee Cyber Security awareness training
- Email Filtering
- Multi Factor (MFA)
- Password management policies
- Regular phishing Simulations

Responsibilities For Employee:

- Verify Suspicious emails
- Avoid Clicking unknown Links
- Report Phishing immediately
- Never Share Passwords.
- Follow Company security policies.

Prevention Measure:

Technical Measures:

- Multi Factor Authentication (MFA)
- Email Security gateways.
- Anti-phishing Filters
- Endpoint protection
- DNS Filtering

Organizational Measures:

- Security awareness programs.
- Password policies
- Regular audits.
- Incident response plans.
- Phishing Simulation exercise.

Conclusion:

Both technology and Employee awareness are essential to reduce phishing attacks...

4) Student Exploits Banking Vulnerability Before Reporting...

A) Analysis:- The Student's intention may have been good, the actions were not lawful or Ethical Because Customer's information without authorization...

Ethical Hacking vs Unauthorized Hacking:-

Ethical Hacking
write permission
Define Scope
Legal activity
Responsible disclosure
Protect Systems

Unauthorized Hacking
No permission
No Approved Scope
Illegal Activity
Unauthorized exploitation
Violates privacy and security...

Relevant IT Act Provisions:-

Section 43: Unauthorized Access to Computer Resources.

Section 66: Criminal offense is dishonest or Fraudulent

Section 72: Breach of Confidentiality and Privacy...

Why Student May Face Legal Action:-

⇒ Accessed Customer's information without authorization

⇒ The Bank has the Right to initiate Legal Proceeding of intent.

Conclusion:-

Good intention do not make unauthorized testing Legally. Ethical Hacking Always authorization Limited testing disclosure For Privacy.